

Geographic Information Retrieval

Lecture 1: Course Introduction — What Makes Search Geographic?

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&

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About me

Yingjing Huang

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Background: GIScience, GeoAI, Urban Data

More details: <https://yingjinghuang.github.io/>

My research interests

01

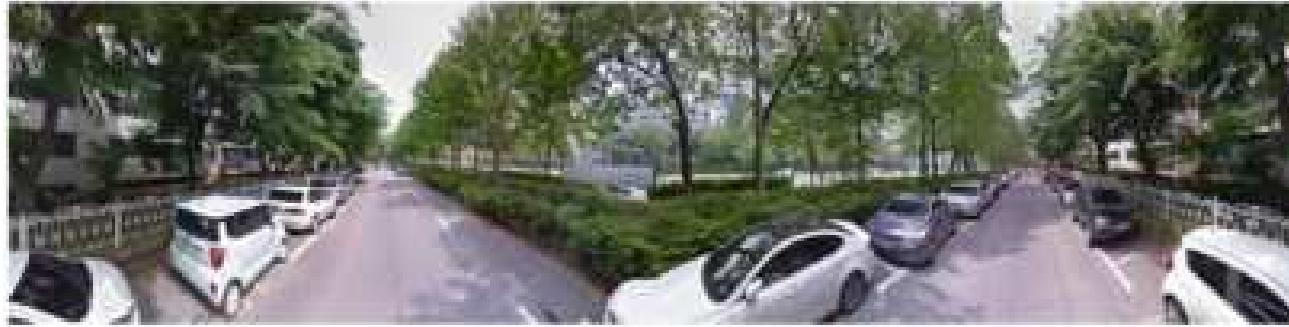
GeoAI for urban problems

02

Representation learning of urban data

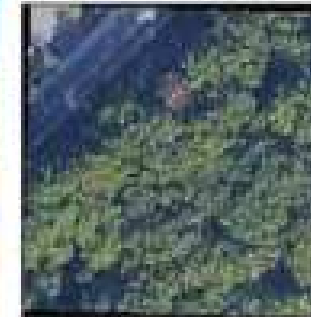
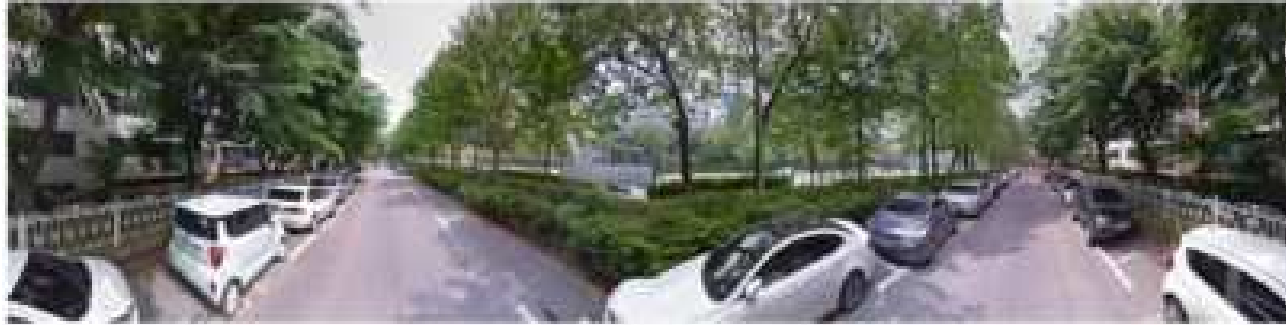
Example 1 : No “true” greenery

Urban greenery measurement exists limitations



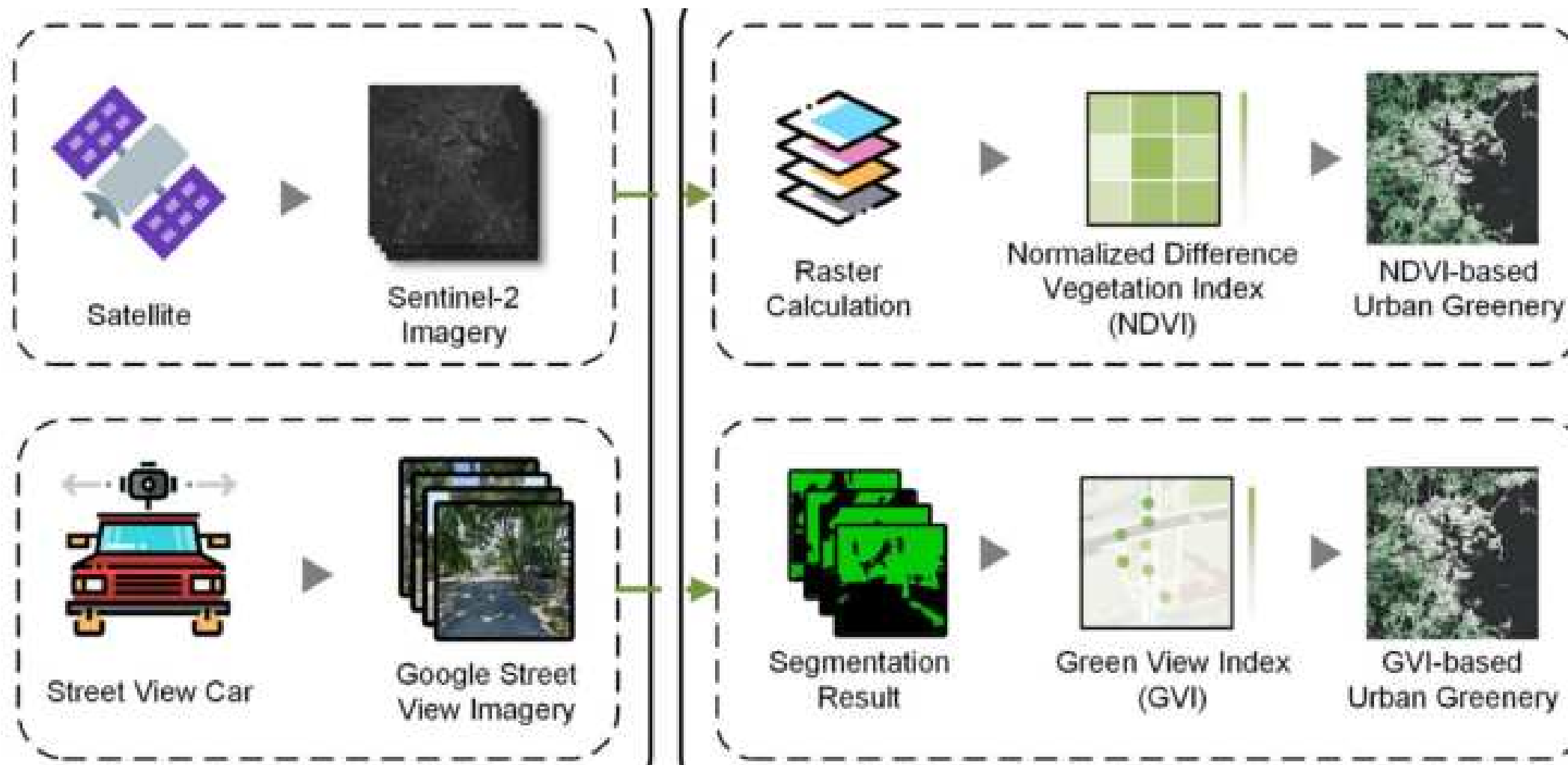
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Urban greenery measurement exists limitations

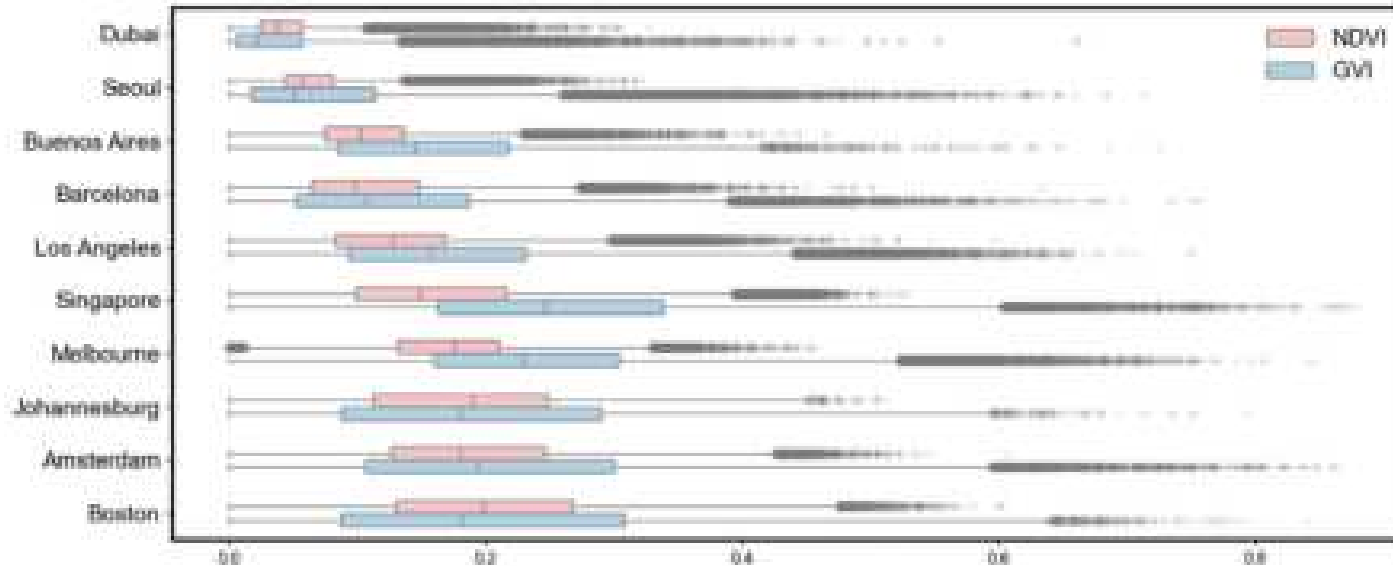


Example 1 : No “true” greenery

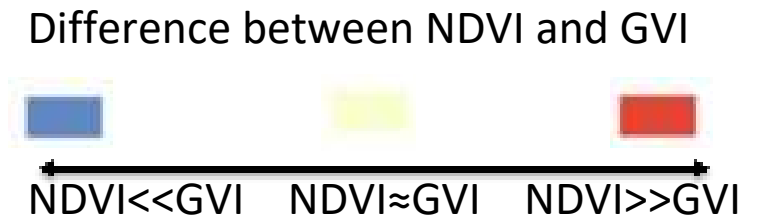
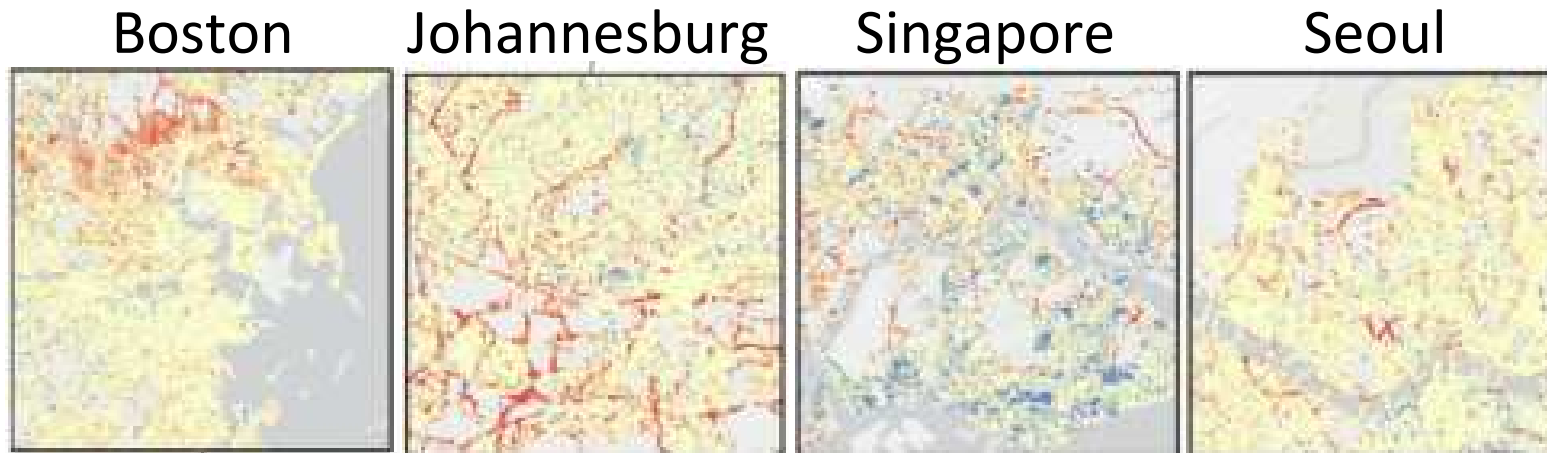
Urban greenery measurement exists limitations



Example 1 : No “true” greenery



Statistical distributions
are similar, **but spatial
distributions differ!**



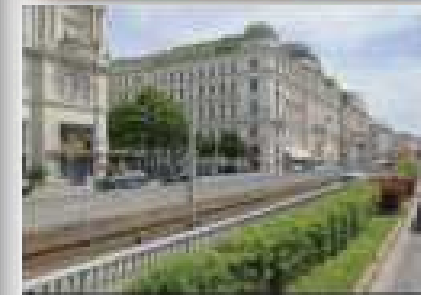
More details: [No “true” greenery paper](#)

Example 2 : Spatiotemporal contrastive learning

Where was this street view image taken?



Example 2 : Spatiotemporal contrastive learning



We design distinct methods to guide computers to focus on the information we need based on different tasks

About me

Zilong Liu

PhD Student

Department of Geography and Regional Research

Background: GIScience

More details: <https://zilongliu-geo.github.io>

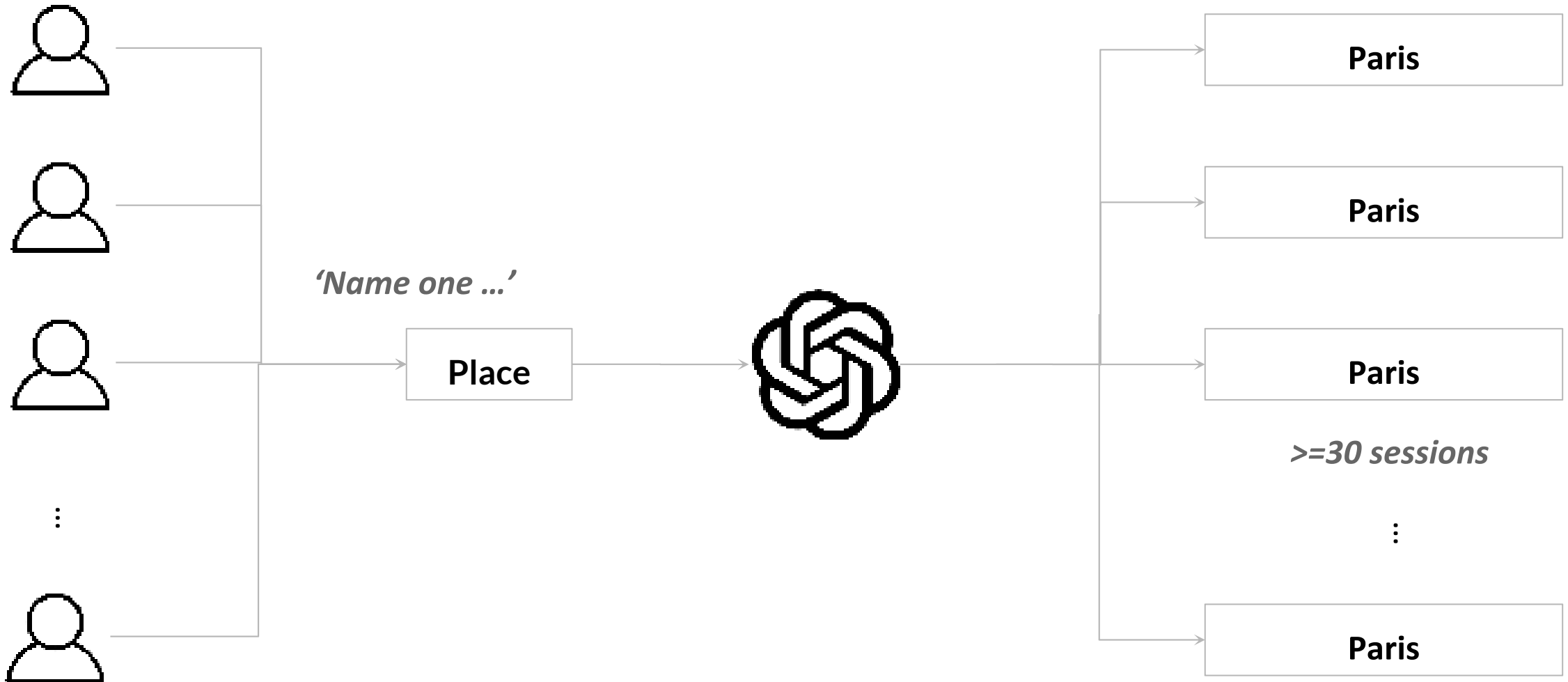
My research interest

01

Geographic Bias in AI Evaluation

Example 1: Paris as Always

Where do GenAI systems favor?

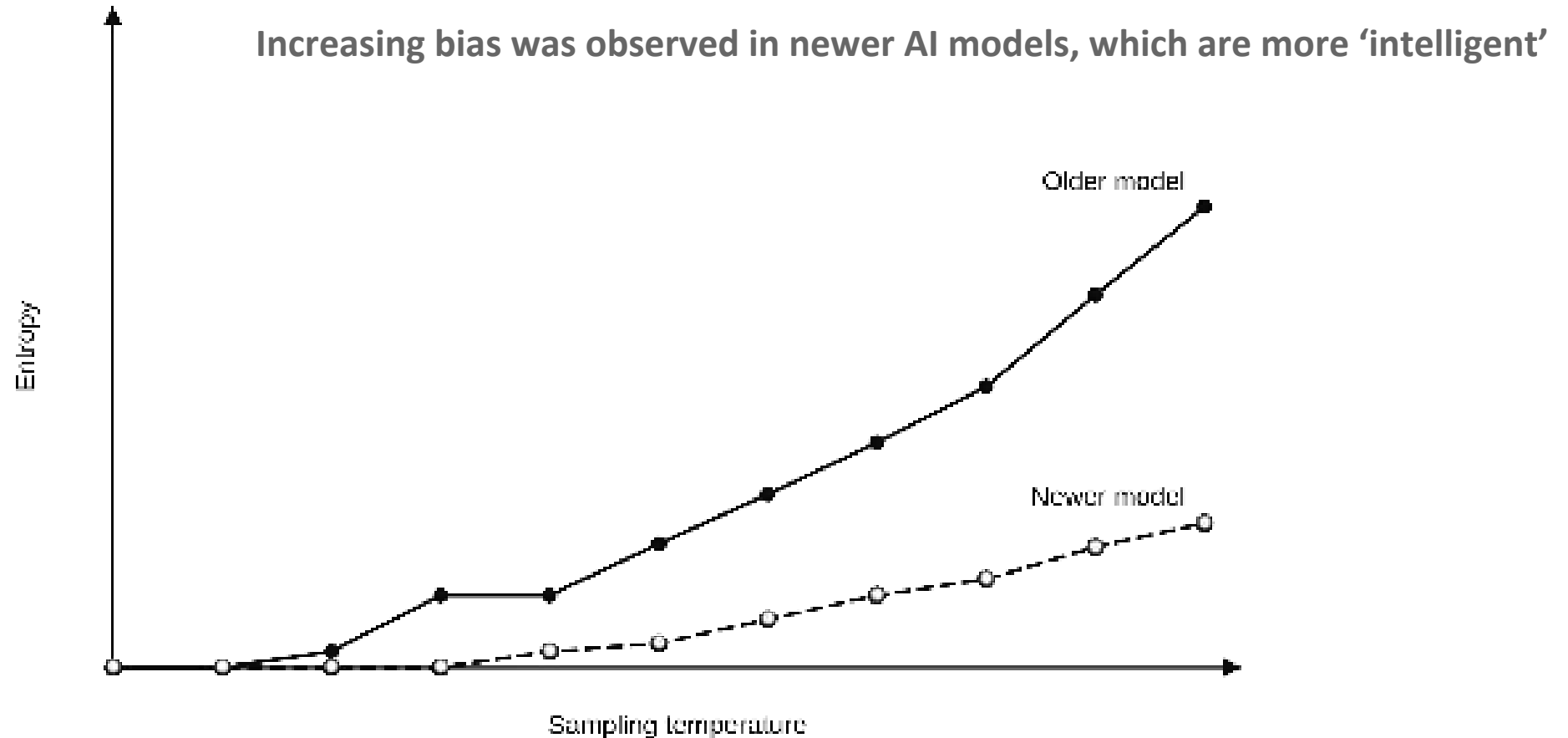


Example 2: Bias Measurement

How terrible is this bias?

Increasing bias corresponds to decreasing entropy

$$\text{Entropy: } H = - \sum p(x) \log p(x)$$



Who are you?

- Name?
- Area of study?
- Master's program?
- What year are you in the program?
- GIS and/or coding experience?
- What do you hope to learn or get out of this course?



Today's plan

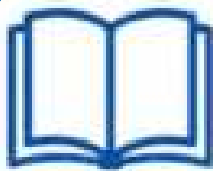
1



Course overview

- Course structure
- Grading
- Course materials

2



Lecture

- Introduction to Information Retrieval (IR)
- Introduction to Geographic Information Retrieval (GIR)
- How IR and GIR are related

3



Lab

- Introduction to Jupyter notebook
- Retrieve a text document from Wikipedia

What is this course about?

“ How search systems retrieve information
when geographic meaning matters ”

- Search is not only about words
- Some queries contain places, regions, distances, directions, and spatial relations
- GIR connects text retrieval with geographic interpretation

Example:

“Find documents about castles **near** Vienna.”



Text Retrieval



Geographic
Interpretation

GIR pipeline



Geoparsing

Geocoding

Text Indexing

Spatial Indexing

Ranking

GIR components

Course roadmap

Week	Group Tue	Group Fri	Topic	Lecturer
Week 1	Oct 6	Oct 9	Introduction	
Week 2	Oct 13	Oct 16	Geoparsing I	Zilong
Week 3	Oct 20	Oct 23	Geoparsing II	Zilong
Week 4	Oct 27	Nov 13	Geocoding I + Quiz (Save the date!)	Zilong
Week 5	Nov 03	Nov 20	Geocoding II	Zilong
Week 6	Nov 10	Nov 27	Midterm exam (Save the date!)	
Week 7	Nov 17	Dec 4	Text Indexing	Yingjing
Week 8	Nov 24	Dec 11	Spatial Indexing + Project proposal (Save the date!)	Yingjing
Week 9	Dec 01	Dec 18	Textual Ranking	Yingjing
Week 10	Dec 15	Jan 8	Spatial and Hybrid Ranking	Yingjing
Week 11	Jan 12	Jan 15	Final project presentation (Save the date!)	

No make-up exam!

How each class works



Lecture

- Main concepts
- Worked examples
- Small methods or code
- ...



Lab

- Walk through part of the notebook
- Understand how the code works
- Practice by yourself
- ...

A lab checkpoint should be submitted every week

Readings and references



- There is **no compulsory textbook**
- **Additional reading materials** will be provided every week

Use the readings to review the main concepts and to see how the lecture topics are discussed in the literature

Resources:

Geographic information retrieval: Progress and Challenges in Spatial Search of Text:
<https://orca.cardiff.ac.uk/id/eprint/110079/1/CBJones-FnTR-GIR-2018.pdf>

Grading

- **Participation (10%)**
 - Submit a lab checkpoint for each class **by 23:59 on the day of the class**
- **Quiz (15%)**
 - Held in Week 4
- **Midterm exam (30%)**
 - Held in Week 6
- **Project proposal (15%)**
 - Detailed instructions will be released in Week 8
 - The proposal should be submitted by each team in Week 9
- **Final project (30%)**
 - Team presentation in Week 11

Course materials

GitHub



- Teaching materials
- Public access

<https://yingjinghuang.github.io/Geographic-Information-Retrieval/>

Moodle



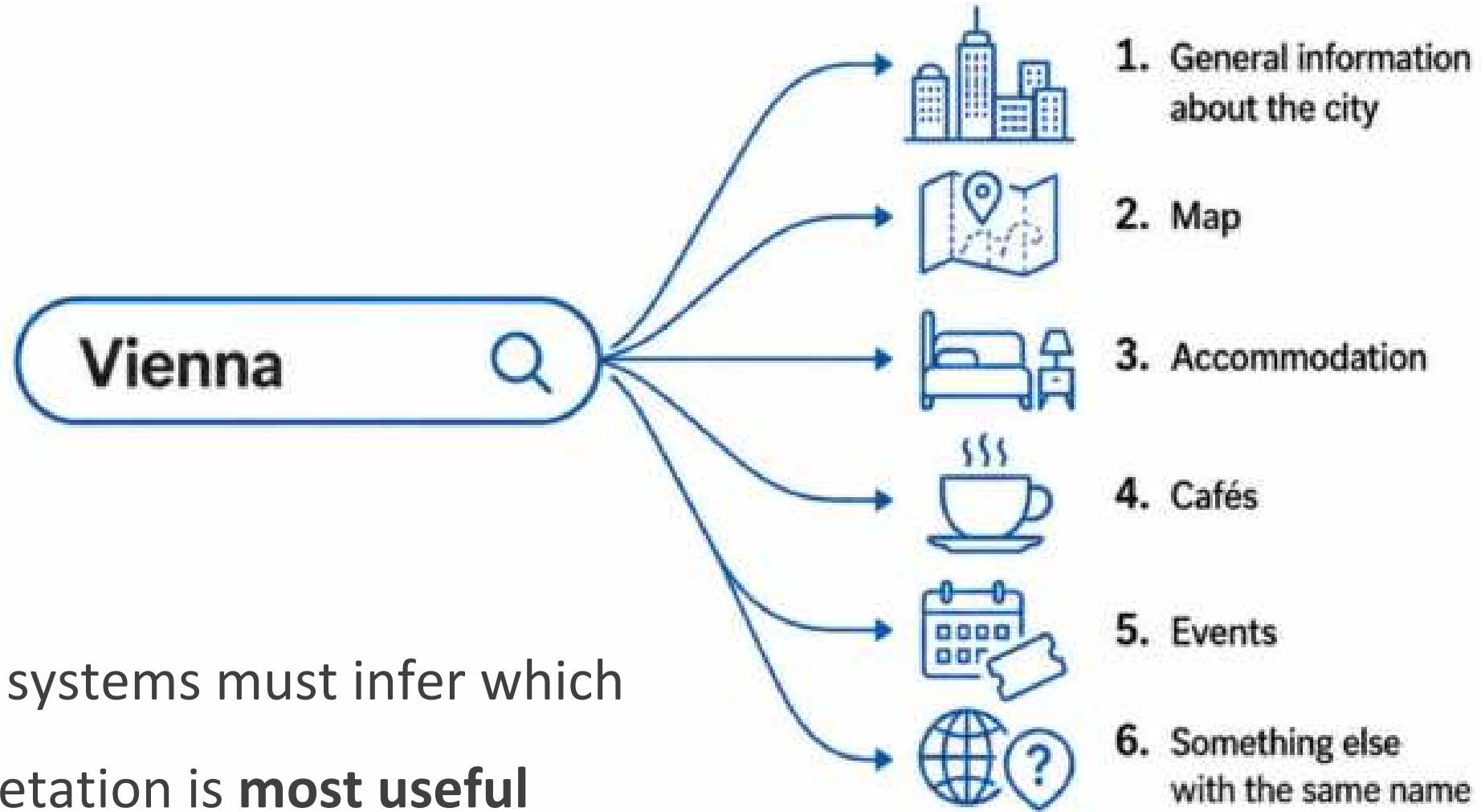
- Teaching materials
- Announcements
- Submissions (e.g., lab checkpoints)
- Grades

From one simple question



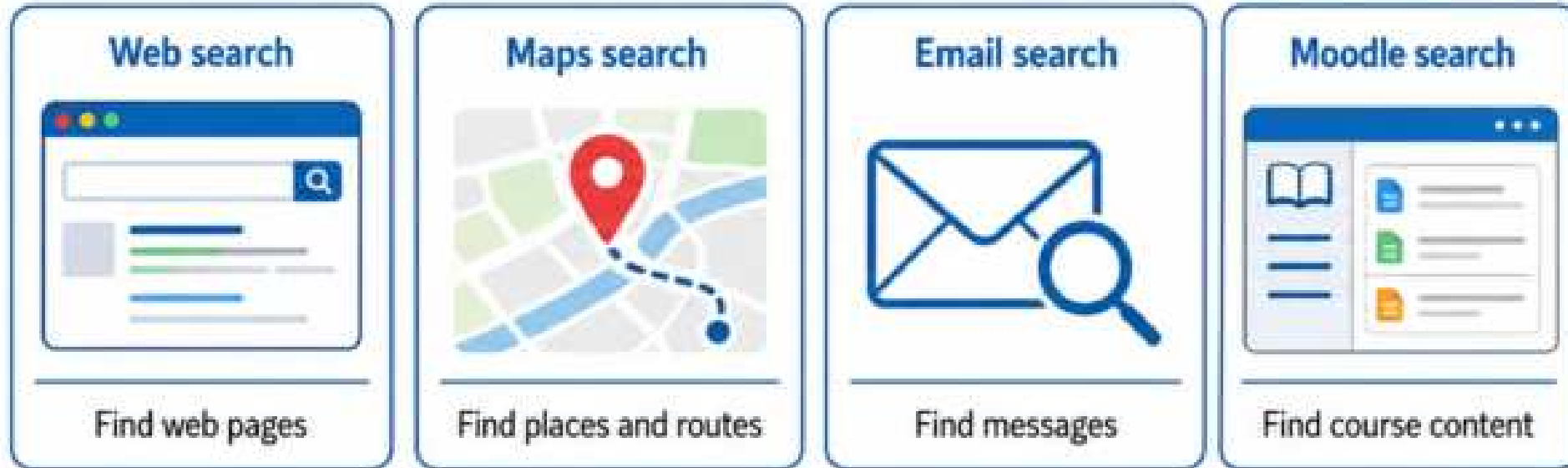
What do you expect to get back?

Possible meanings of the query



Search systems must infer which interpretation is **most useful**

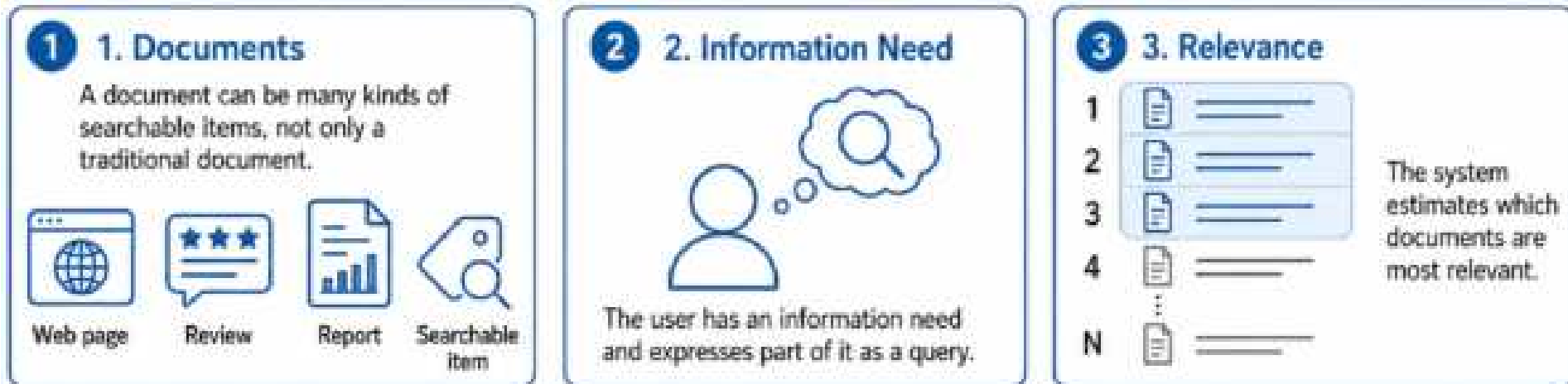
Search in everyday life



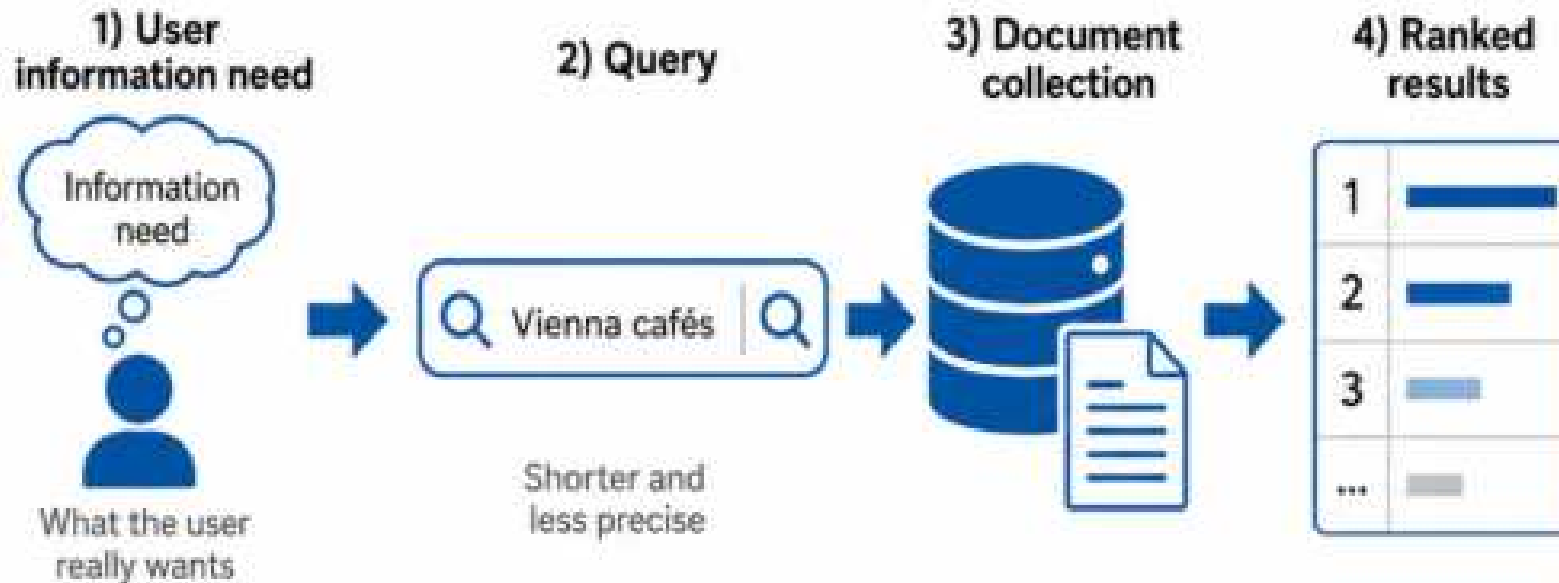
Different systems search different things,
But the core retrieval idea is the same

What is Information Retrieval?

“ Information Retrieval is the process of finding and ranking information that is relevant to a user’s information need.”



Documents, queries, and relevance



Why ranking matters?

- Relevance is not simply yes or no
- Some documents are more useful than others
- The system estimates usefulness under uncertainty

IR is not database search



Database search

```
SELECT * FROM cafés  
WHERE city = 'Vienna'
```

Name	City	Type
Café Central	Vienna	Café
Balthasar	Vienna	Café
Café Landtmann	Vienna	Café
Café Sperl	Vienna	Café
Café Sacher	Vienna	Café
Café Mozart	Salzburg	Café
Café Einstein	Graz	Café

- ✓ Known fields
- ✓ Explicit conditions
- ✓ Clear match / no match

from exact
filtering to
ranked
relevance



Information Retrieval

quiet cafés near the University of Vienna



quiet?

near?



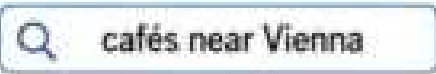
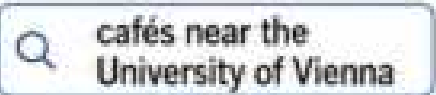
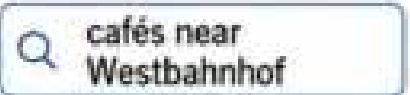
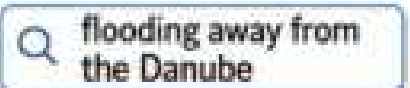
- 1 Café Central
- 2 Café Landtmann
- 3 Balthasar

- ✓ Vague user need
- ✓ Partly relevant results
- ✓ Ranking is needed

When does search become geographic?



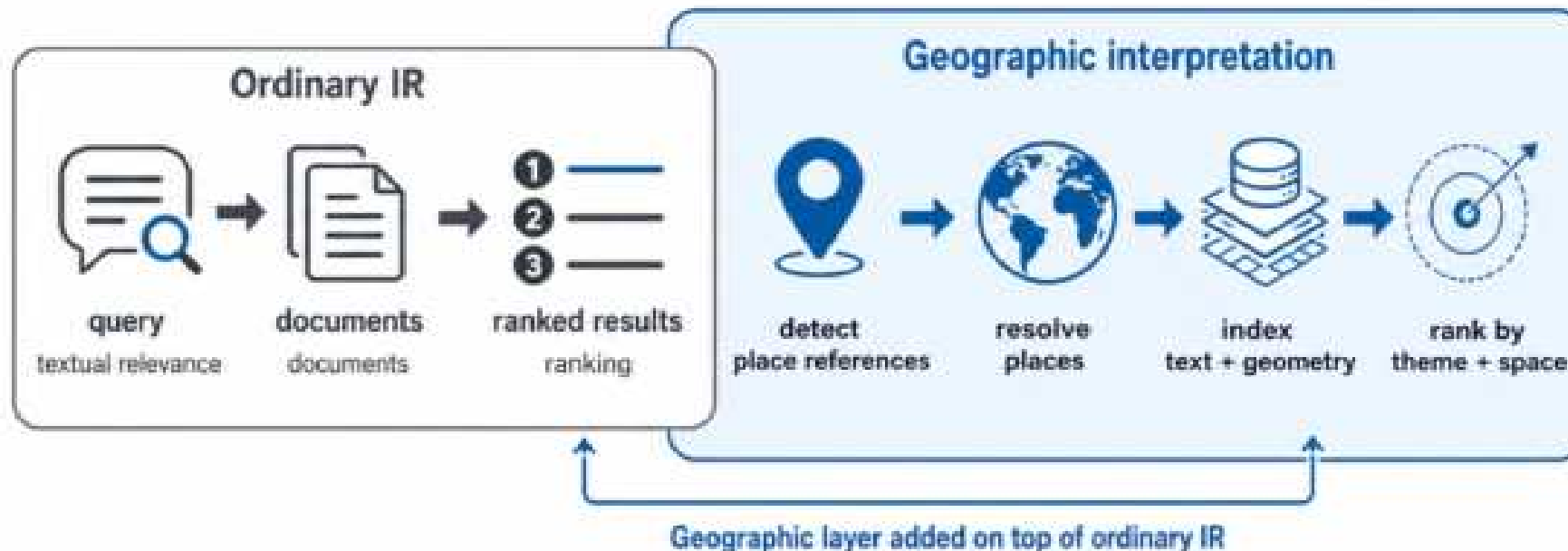
When does search become geographic?

Place changes	  →  	 Different candidate documents and places
Reference location changes	  →  	 Different local search area
Spatial relation changes	  →  	 Different geographic meaning

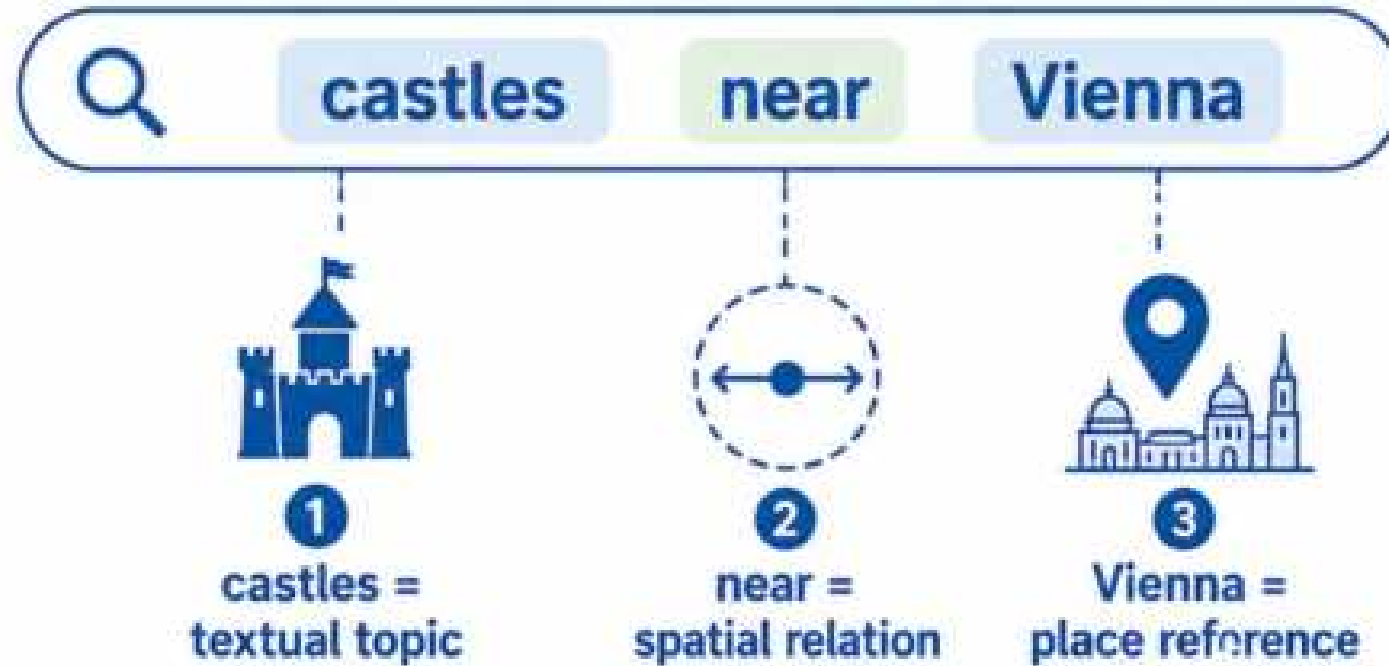
**Geography changes what the system retrieves
and how results should be ranked**

What is Geographic Information Retrieval?

“ Geographic Information Retrieval is Information Retrieval where **geographic meaning** affects retrieval. ”

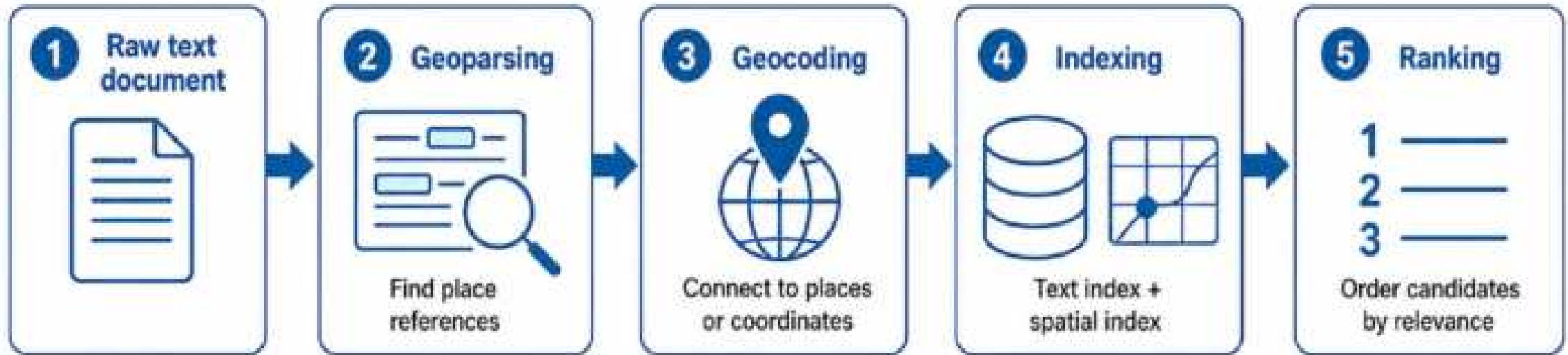


Example: “castles near Vienna”



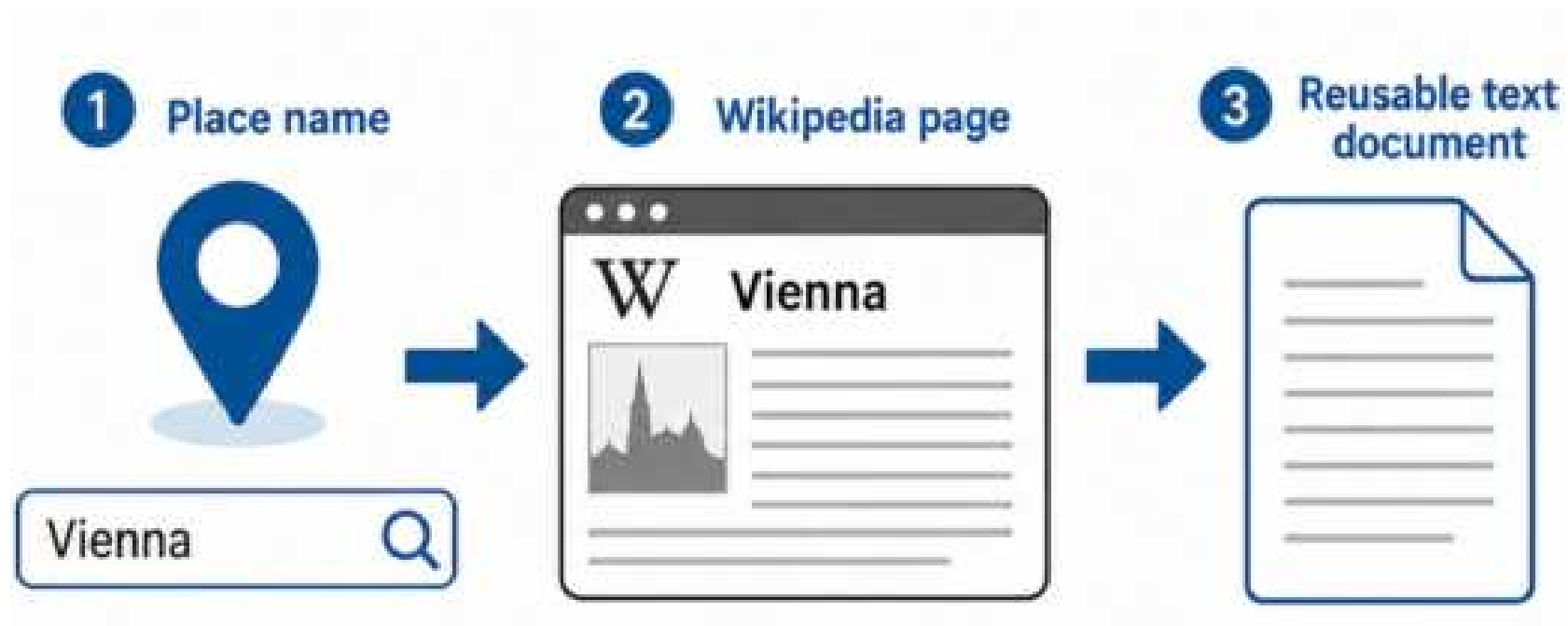
A short query can contain **topic**, **place**, and **spatial relation**

A GIR system needs several components



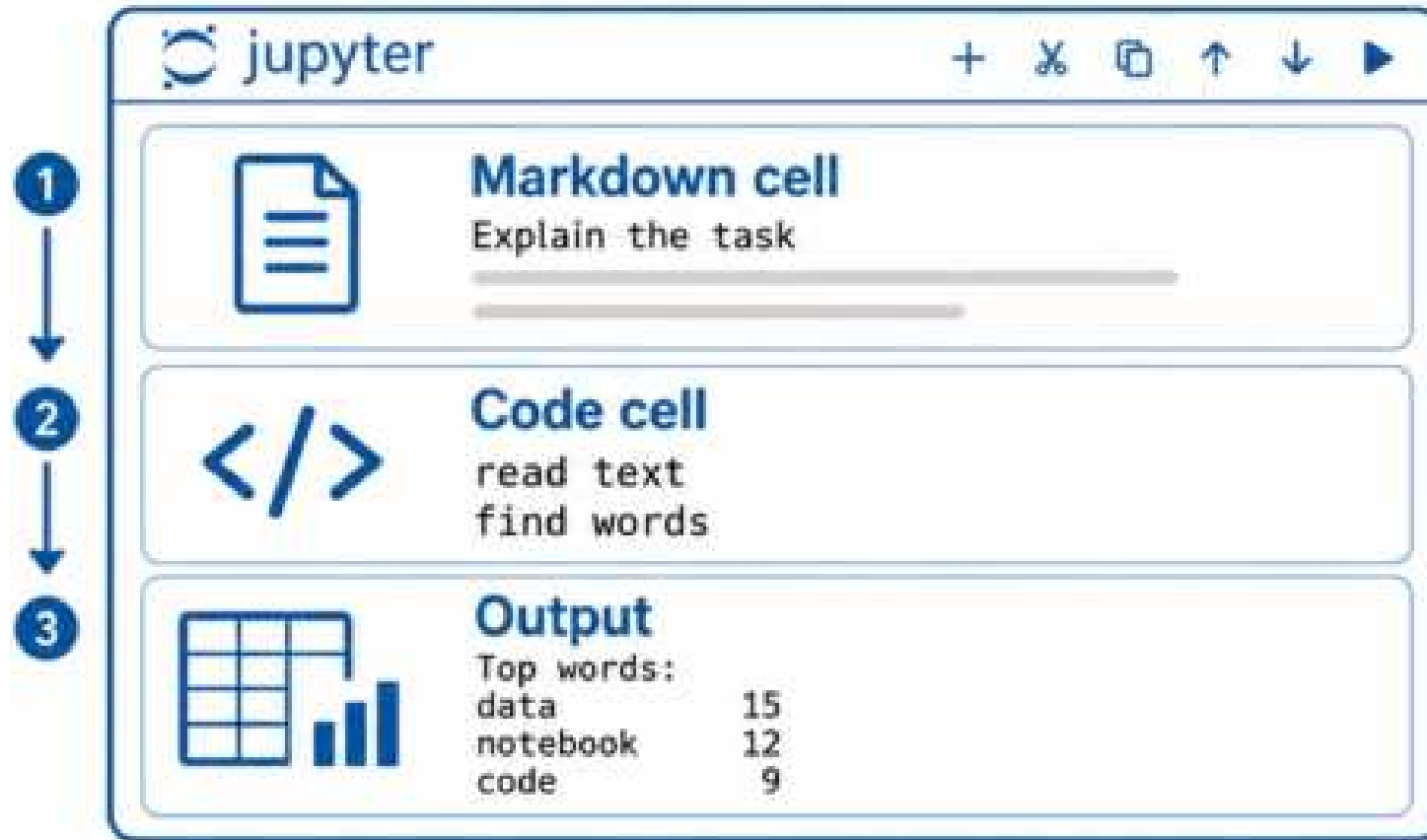
First lab: retrieving a text document

Use a place name to retrieve a Wikipedia page
and treat it as a text document

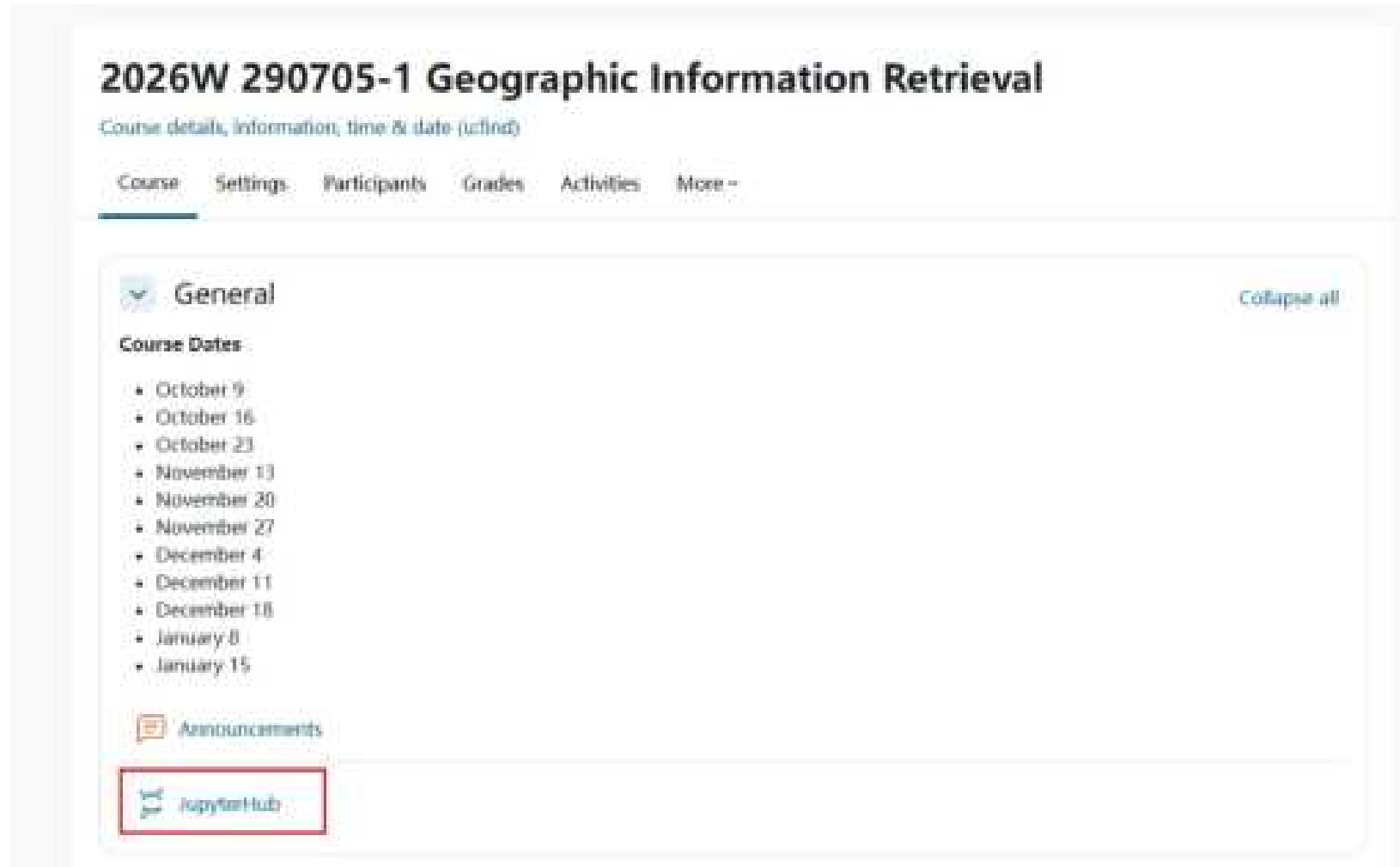


What is a Jupyter notebook?

A notebook keeps **explanation**, **code**, and **results** together



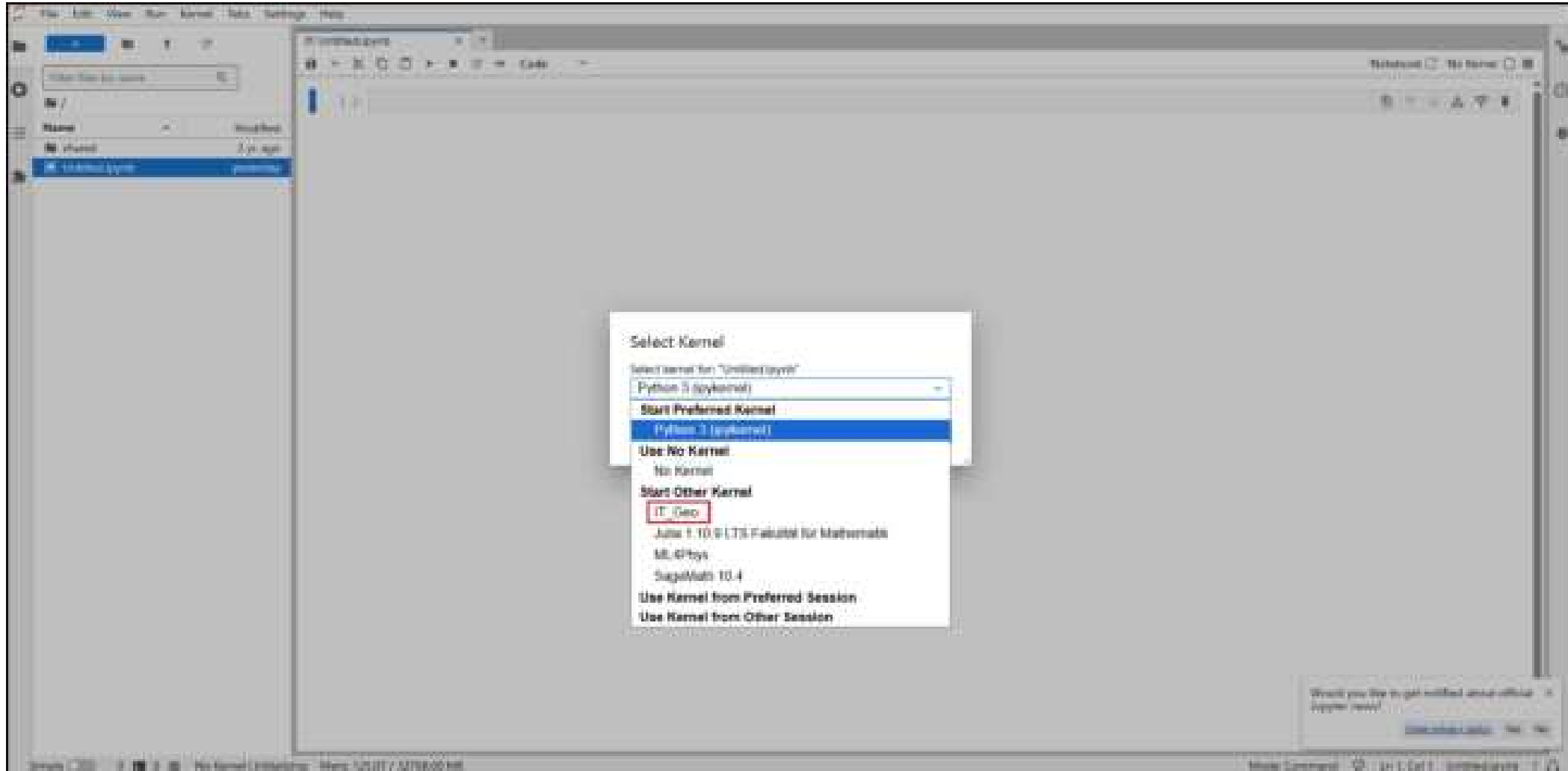
How to use notebooks in this course



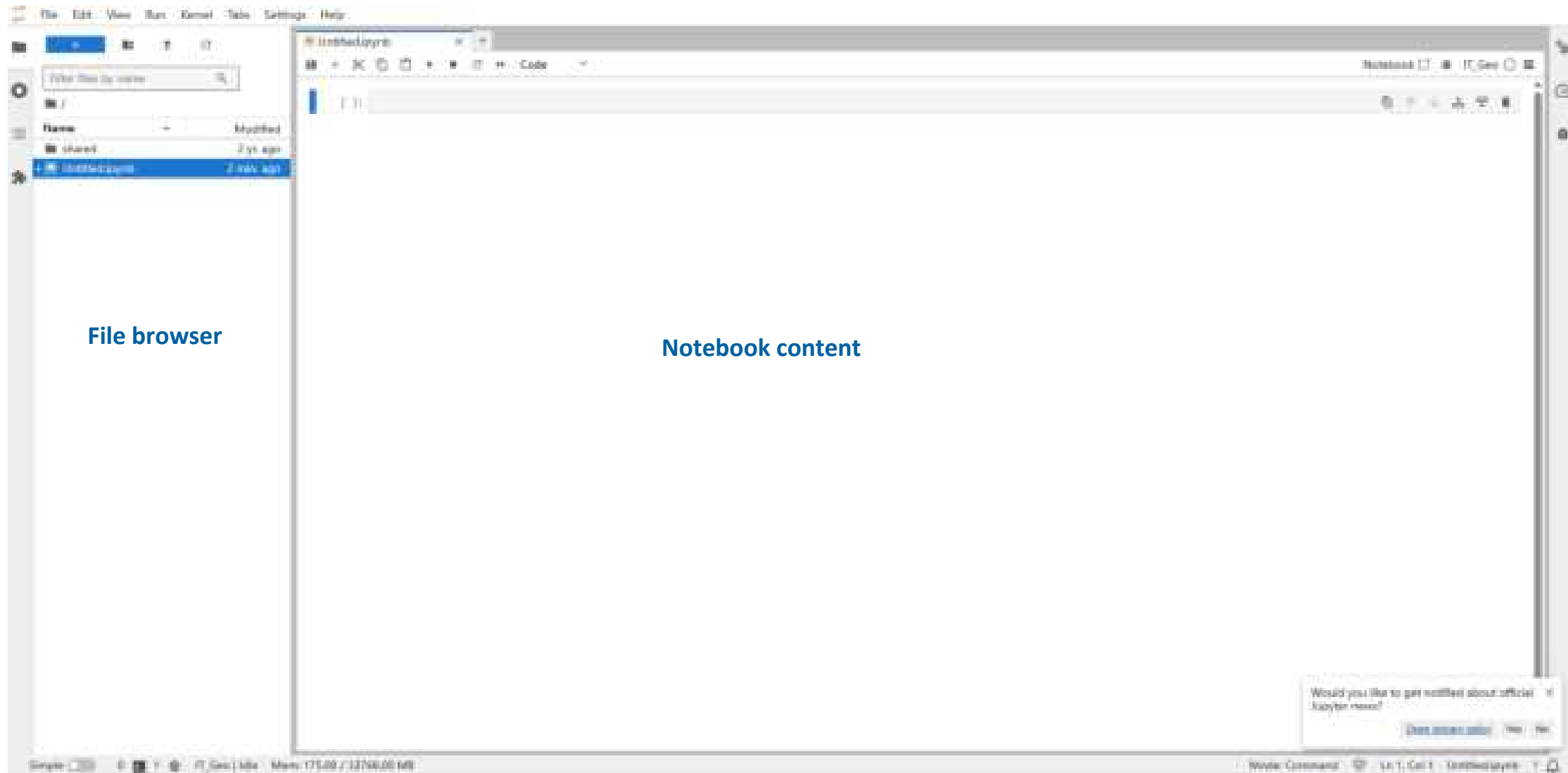
How to use notebooks in this course



How to use notebooks in this course



How to use notebooks in this course



Lab time

*Please remember to submit your lab checkpoint
by the end of the day.*